

Regression Analysis (I)

Midterm Exam.

April 19, 2021

- 1.(40) Let $\mathbf{A}$ be $n \times p$ full column matrix ($p < n$) with singular values $s_1, \dots, s_p$. Compute $\|\mathbf{A}\|_F$, the Frobenius norm of $\mathbf{A}$.
- 2.(40) Let $\mathbf{A}$ be 3×3 matrix with diagonal elements 2, 1, -3 . If two eigenvalues of $\mathbf{A}$ are 1 and 3, find the determinant of $\mathbf{A}$.
- 3.(40) State the difference between the FWER (family-wise error rate) and the FDR (false discovery rate) in multiple hypotheses testing.
- 4.(40) Compute the variance of t -distribution with d.f. $r > 2$.
- 5.(40) Let $\hat{\theta}$ be an estimator of θ . Show that $E[(\hat{\theta} - \theta)^2] = Var(\hat{\theta}) + Bias(\hat{\theta})^2$.
- 6.(80) Let $X_1, \dots, X_n$ be r.s. from a Poisson distribution with mean λ .
 - (1)(40) Find the MLE of $e^{-\lambda}$.
 - (2)(40) Find a function $g(\cdot)$ when $\sqrt{n}(g(\bar{X}) - g(\lambda)) \rightarrow N(0, 1)$ in distribution.
- 7.(120) Consider a simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$, $i = 1, \dots, n$. Assume that ϵ_i 's are iid $N(0, \sigma^2)$. Let $\hat{\beta}_0$ and $\hat{\beta}_1$ be LSEs of β_0 and β_1 , respectively.
 - (1)(40) Compute w_i , where $\hat{\beta}_0 = \sum_{i=1}^n w_i Y_i$.
 - (2)(40) Compute $E(\hat{\beta}_0)$ and $Var(\hat{\beta}_0)$ using the result in (1).
 - (3)(40) Using the results in (1) and (2), derive $100 \times (1 - \alpha)\%$ confidence interval for β_0 when σ^2 is known.